NMR Spectroscopy

Answer key with worked solutions

A. Environments

1. Four kinds of information.

> (1) **Number of signals**: number of H environments. (2) **Chemical shift δ**: the type of environment. (3) **Integration**: relative number of H in each environment. (4) **Splitting**: number of H on neighbouring carbons.

2. Signals.

> (a) **1** (all 6 H equivalent) (b) **2**, ratio **6 : 2** (= 3 : 1) (c) **2**, ratio **9 : 1** (d) **1** (all 6 H equivalent) (e) **3**: CH_3 of ethyl, CH_2, and O–CH_3, ratio **3 : 2 : 3**

B. Chemical shift

3. Shift calculations.

> (a) $\delta = \dfrac{2220\ \text{Hz}}{600\ \text{MHz}} = \mathbf{3.70\ ppm}$
> (b) $1.20\ \text{ppm} \times 300\ \text{MHz} = \mathbf{360\ Hz}$ from TMS

C. Splitting and predicting spectra

4. Splitting.

> (a) CH_2 (next to 3 H): **quartet**, about 3.6 ppm; CH_3 (next to 2 H): **triplet**, about 1.5 ppm.
> (b) CH (next to 6 equivalent H): **septet** (7 peaks); each CH_3 (next to 1 H): **doublet**, area 6.

5. Propanoic acid.

> | Group | δ (ppm) | Area | Splitting |
> |---|---|---|---|
> | CH_3 | about 1.2 | 3 | triplet |
> | CH_2 (next to C=O) | about 2.4 | 2 | quartet |
> | COOH | about 11–12 | 1 | singlet |

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6. ^{13}C signals.

> (a) **2** (the two CH_3 are equivalent; plus C=O) (b) **4** (CH_3 next to C=O, C=O, O–CH_2, CH_2–CH_3) (c) **1** (all six carbons equivalent)

D. Solving structures

7. C_3H_8O.

> **Propan-2-ol**, $(CH_3)_2CHOH$. Doublet (6H) at 1.2 ppm: two equivalent CH_3 groups next to one H. Septet (1H) at 4.0 ppm: the CH on the carbon bonded to O, next to 6 H. Singlet (1H) at 2.2 ppm: the OH.

E. Challenge

8. C_4H_8O.

> **Butanone**, $CH_3COCH_2CH_3$. IR 1715 cm^{-1}: **C=O** (no O–H). MS m/z 72: molar mass of C_4H_8O. NMR: singlet (3H) at 2.1 ppm: CH_3 next to C=O with no neighbouring H; quartet (2H) at 2.4 ppm: CH_2 next to C=O and next to a CH_3; triplet (3H) at 1.1 ppm: the CH_3 of the ethyl group, next to CH_2.

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